



Theory of Errors

Multivariate Error
Propagation

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4

Lesson objectives

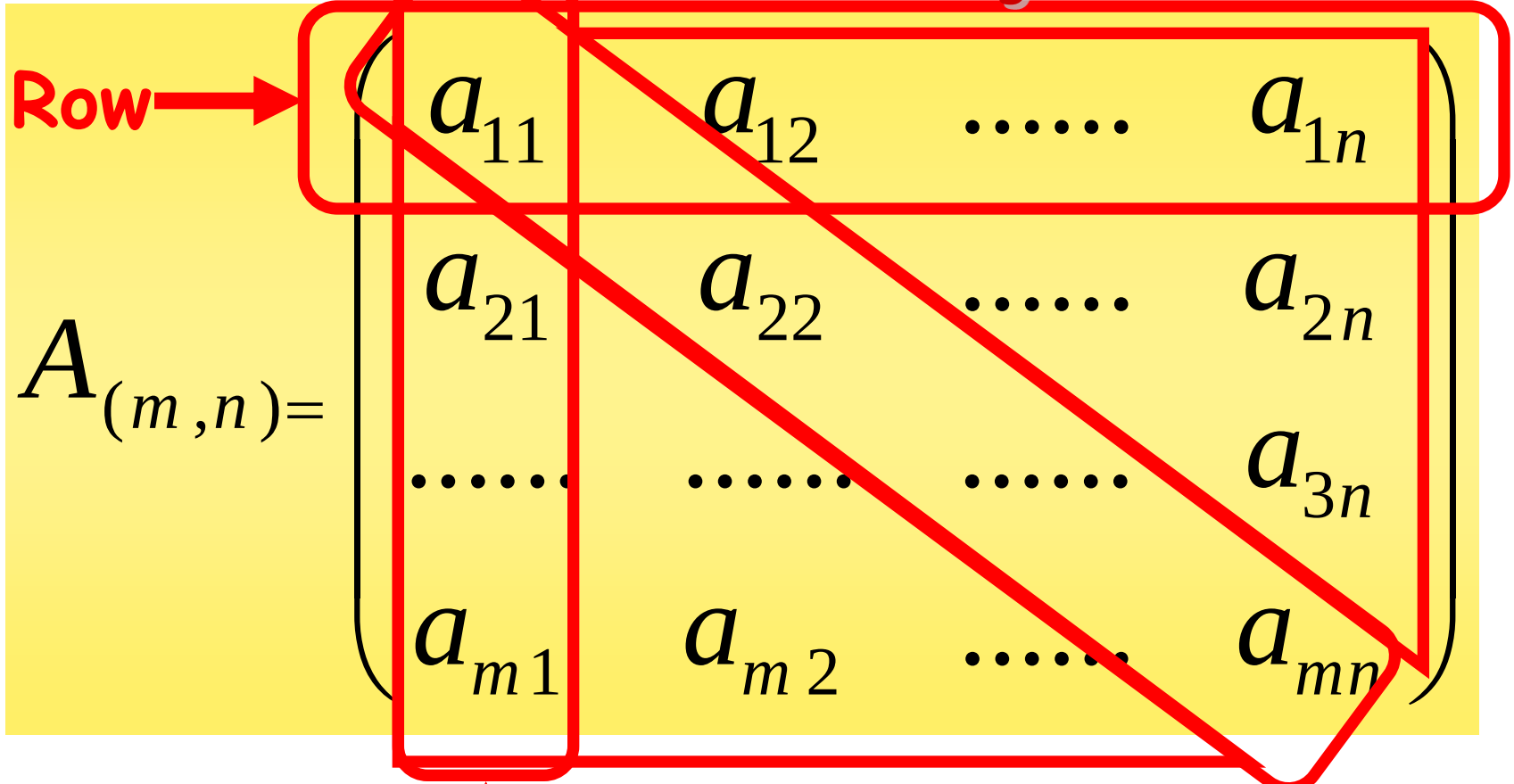
- Matrix algebra background.
- Error propagation of single unknown as a function of multiple uncorrelated observations.
- Error propagation of multiple unknown as a function of multiple correlated observations.

Matrix algebra background

- A matrix is a rectangular array of elements.
- Each matrix has rows and columns and this defines the size of the matrix.
- If a matrix A has m rows and n columns, the size of the matrix is denoted by $m \times n$.

Matrix algebra background

Off Diagonal elements



Column

Diagonal elements

Special types of matrices

- **Vector:** A vector is a matrix that has only one row or one column.
- There are two types of vectors :
 - **Row vector:** If a matrix has one row, it is called a row vector and 'm' is the dimension of the row vector.

$$B_m = (b_1 \quad b_2 \quad \dots \quad b_m)$$

Special types of matrices

- Column vector: If a matrix has one column, it is called a column vector and n is the dimension of the vector.

$$C_n = \begin{pmatrix} c_1 \\ c_1 \\ \dots \\ c_n \end{pmatrix}$$

Square matrix

- **Square matrix:** If the number of rows (m) of a matrix is equal to the number of columns (n) of the matrix, ($m = n$), it is called a square matrix.

Square matrix

$$A_{(n,n)} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & a_{3n} \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Upper triangular matrix

- **Upper triangular matrix:** A $m \times n$ matrix for which $a_{ij} = 0$, $i < j$ is called an upper triangular matrix. That is, all the elements below the diagonal entries are zero.

Upper triangular matrix

$$A_{(4,4)} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{pmatrix}$$

Lower triangular matrix

- **Lower triangular matrix:** A $m \times n$ matrix for which $a_{ij} = 0, j > i$ is called a lower triangular matrix. That is, all the elements above the diagonal entries are zero.

Lower triangular matrix

$$A_{(4,4)} = \begin{pmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Law of propagation of V-C matrix

- Single unknown X .
- Multiple uncorrelated observations $F(L)$.

$$X = f(l_1, l_2, l_3, \dots, l_n) = f(L)$$

Variance of computed value x

$$\sigma_x^2 = \left(\frac{\partial x}{\partial l_1}\right)^2 \sigma_1^2 + \left(\frac{\partial x}{\partial l_2}\right)^2 \sigma_2^2 + \dots + \left(\frac{\partial x}{\partial l_n}\right)^2 \sigma_n^2$$

$$\sigma_x^2 = \left(\frac{\partial x}{\partial l_1}\right) \sigma_1^2 \left(\frac{\partial x}{\partial l_1}\right) + \left(\frac{\partial x}{\partial l_2}\right) \sigma_2^2 \left(\frac{\partial x}{\partial l_2}\right) + \dots + \left(\frac{\partial x}{\partial l_n}\right) \sigma_n^2 \left(\frac{\partial x}{\partial l_n}\right)$$

Variance of computed value x in matrix form

$$\sigma_x^2 = \begin{pmatrix} \frac{\partial x}{\partial l_1} & \frac{\partial x}{\partial l_2} & \dots & \frac{\partial x}{\partial l_n} \end{pmatrix} \begin{pmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 \\ 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \sigma_n^2 \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial l_1} \\ \frac{\partial x}{\partial l_2} \\ \dots \\ \frac{\partial x}{\partial l_n} \end{pmatrix}$$

Variance of computed value x in matrix form

$$\sigma^2_{x(1,1)} = J_{L(1,n)} \Sigma_{L(n,n)} J^T_{L(n,1)}$$

Where :

J : all partial derivatives of the unknown
w.r.t. all observed quantities.

Σ : Variances matrix.

Law of propagation of V-C matrix

- Multiple unknown vector X .
- Multiple correlated observations $F(L)$.

$$X_{u,1} = f_{u,1}(L_{n,1})$$

$$X_1 = f_1(l_1, l_2, \dots, l_n)$$

$$X_2 = f_2(l_1, l_2, \dots, l_n)$$

.....

$$X_u = f_u(l_1, l_2, \dots, l_n)$$

V-C matrix of computed values

$$\Sigma_{x(u,u)} = J_{L(u,n)} \Sigma_{L(n,n)} J^T_{L(n,u)}$$

Where :

- $\Sigma_{x(u,u)}$: V-C matrix of unknowns.
- J : all partial derivatives of all unknowns w.r.t.all observed quantities "Jacobian matrix" and called coefficient matrix in case of linear mathematical model.
- $\Sigma_{L(n,n)}$: V-C matrix of observations.

$\Sigma_{x(u,u)}$ V-C matrix of unknowns

$$\Sigma_{x(u,u)} = \begin{pmatrix} \sigma_{x_1}^2 & \sigma_{x_1 x_2} & \sigma_{x_1 x_3} & \dots & \sigma_{x_1 x_u} \\ & \sigma_{x_2}^2 & \sigma_{x_2 x_3} & \dots & \sigma_{x_2 x_u} \\ & & \dots & \dots & \dots \\ & & & \dots & \dots \\ & & & & \sigma_{x_u}^2 \end{pmatrix}$$

Symmetric

$J_{L(u,n)}$ Jacobian matrix

$$J_{L_{u,n}} = \begin{pmatrix} \frac{\partial x_1}{\partial l_1} & \frac{\partial x_1}{\partial l_2} & \dots & \dots & \frac{\partial x_1}{\partial l_n} \\ \frac{\partial x_2}{\partial l_1} & \frac{\partial x_2}{\partial l_2} & \dots & \dots & \frac{\partial x_2}{\partial l_n} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial x_u}{\partial l_1} & \frac{\partial x_u}{\partial l_2} & \dots & \dots & \frac{\partial x_u}{\partial l_n} \end{pmatrix}$$

$\Sigma_{L(n,n)}$ V-C matrix of observations

$$\Sigma_{L(n,n)} = \begin{pmatrix} \sigma_{l_1}^2 & \sigma_{l_1 l_2} & \sigma_{l_1 l_3} & \dots & \sigma_{l_1 l_u} \\ & \sigma_{l_2}^2 & \sigma_{l_2 l_3} & \dots & \sigma_{l_2 l_u} \\ & & \dots & \dots & \dots \\ & \text{Symmetric} & & \dots & \dots \\ & & & & \sigma_{l_n}^2 \end{pmatrix}$$

Summary of steps for solving a multivariate error propagation problem

- Here we have number of observables, each given with its mean value and the standard deviation of the mean with any additional information concerning the existing correlation among them

$$\text{Given: } \bar{L}_{(n,1)}, \sum \bar{L}_{n,n}$$

$$\text{Required: } \hat{X}_{(u,1)}, \sum \hat{X}_{(u,u)}$$

Step 1

- Identifying the components of observation vector L and unknown vector X .
- Forming the explicit mathematical model “one equation must be formulated for each single unknown”

$$X_{(u,1)} = F_{(u,1)}(L_{(n,1)})$$

Step 2

- Establish the V-C matrix of observed quantities by:
 - Squaring the standard deviations and put them on the main diagonal.
 - Calculating the covariance for any two correlated observations and put it in the appropriate off-diagonal position.

Keep in mind the units

Step 3

- Evaluate the elements of jacobian matrix by:
 - Performing the partial derivatives of the unknowns w.r.t. observations.
 - Substituting with the numerical values of the observations.

$$\sum_{\hat{x}} = J_L \sum_{\bar{L}} J^T_L$$

Step 4

- Adjust the physical units in the covariance law.
 - By making sure that the physical units of the elements of jacobian matrix and V-C matrix. "Scaling process"

Step 5

- Transpose jacobian matrix after scaling process.
- Applying the covariance law to compute the variances and covariance of unknowns.

Step 6

- Substituting in the explicit mathematical model for L by the given mean value to obtain the estimate values $\hat{X}$

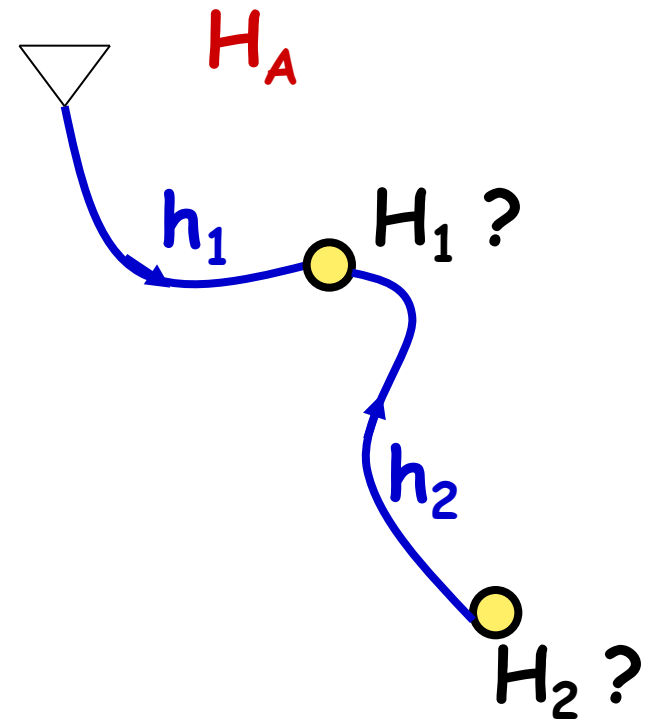
Example 1

Given:

- $H_A = 20\text{m}$ fixed point
- $h_1 = 5\text{ m}$, $\sigma_1 = 2\text{ mm}$
- $h_2 = 8\text{ m}$, $\sigma_2 = 3\text{ mm}$

Required:

- H_1, H_2
- σ_{H_1} , σ_{H_2}



Solution

- **Step 1:**

$$X_{(2,1)} = \begin{pmatrix} H_1 \\ H_2 \end{pmatrix}$$

$$L_{(2,1)} = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

- **The math. model:**

$$H_1 = H_A + h_1 = 20 + 5 = 25m$$

$$H_2 = H_A + h_1 - h_2 = 20 + 5 - 8 = 17m$$

Solution

- Step 2:

Here, we have uncorrelated observations.
i.e. $\sigma_{12} = 0$

- The V-C matrix Σ_L is:

$$\Sigma_{L(2,2)} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$$

Solution

- Step 3:

Here, the math.

Model is linear and hence J_L will be just the coefficient of observations

$$H_1 = H_A + h_1$$

$$H_2 = H_A + h_1 - h_2$$

- The Jacobian matrix J_L is:

$$J_{L(2,2)} = \begin{pmatrix} \frac{\partial H_1}{\partial h_1} & \frac{\partial H_1}{\partial h_2} \\ \frac{\partial H_2}{\partial h_1} & \frac{\partial H_2}{\partial h_2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

Solution

- Step 4:
 - No scaling is needed here.
 - As partial derivatives are unit less,
 - variances observations are in mm^2 which are the same units required for $V-C$ matrix of unknown elements.

Solution

- **Step 5:**
Applying the covariance law to get Σ_x

$$X_{(2,1)} = \begin{pmatrix} H_1 \\ H_2 \end{pmatrix} \quad \Sigma_{X_{(2,2)}} = \begin{pmatrix} \sigma_{H_1}^2 & \sigma_{H_1 H_2} \\ \sigma_{H_2 H_1} & \sigma_{H_2}^2 \end{pmatrix}$$

$$\Sigma_{x_{(2,2)}} = J_{L(2,2)} \Sigma_{\bar{L}(2,2)} J_{L(2,2)}^T$$

Solution

- Step 5:

$$\Sigma_{X_{(2,2)}} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

$$\Sigma_{X_{(2,2)}} = \begin{pmatrix} 4 & 0 \\ 4 & -9 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 13 \end{pmatrix}$$

Solution

- Step 5:

$$\sigma_{H_1} = \sqrt{4} = 2mm$$

$$\sigma_{H_2} = \sqrt{13} = 3.6mm$$

$$\ell_{H_1H_2} = \frac{\sigma_{H_1H_2}}{\sigma_{H_1}\sigma_{H_2}} = \frac{4}{(2)(3.6)} = +0.55$$

Significant

Solution

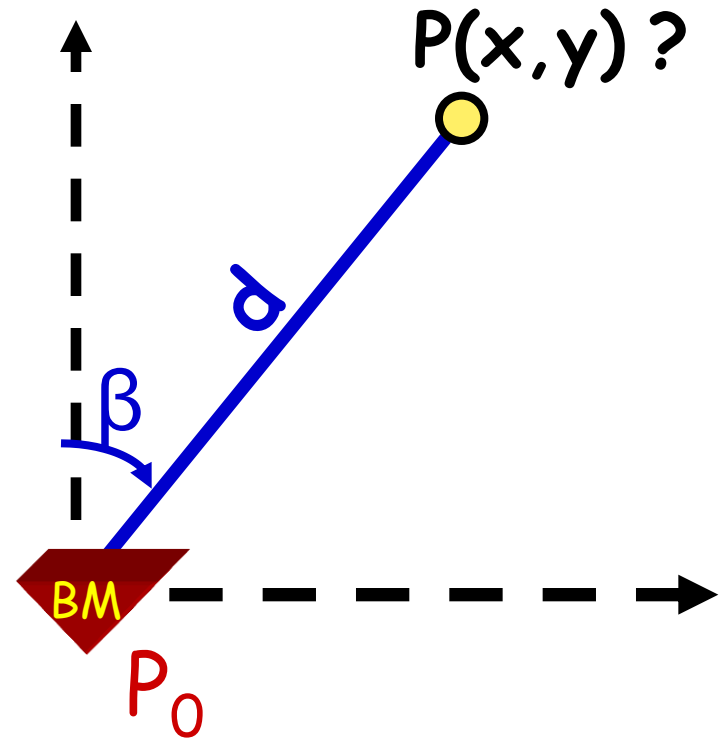
$$H_1 = 25m \pm 2mm$$

$$H_2 = 17m \pm 3.6mm$$

Example 2

Given:

- P_0 is a fixed point
- $x_0 = 500 \text{ m}$, $y_0 = 700 \text{ m}$
- $d = 200 \text{ m}$, $\sigma_d = 2 \text{ cm}$
- $\beta = 30^\circ$, $\sigma_\beta = 3''$
- Required:
- x, y
- σ_x , σ_y



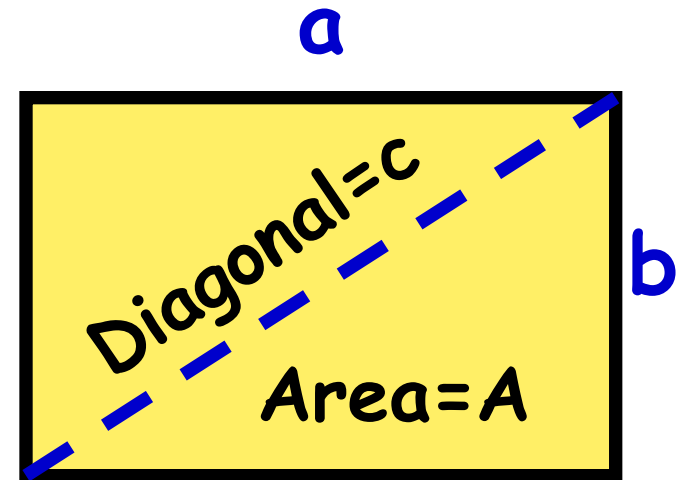
Example 3

Given:

- $a=400$ m , $b=300$ m
- Relative error in $a,b = 10\text{ppm}$
- Correlation between $a,b = 50\%$

• Required:

- Area (A) , Diagonal (c)
- σ_A , σ_c



Solution

- **Step 1:**

$$X_{(2,1)} = \begin{pmatrix} A \\ c \end{pmatrix}$$

$$L_{(2,1)} = \begin{pmatrix} a \\ b \end{pmatrix}$$

- **The math. model:**

$$A = a \cdot b = (400) \cdot (300) = 120000m^2$$

$$c = \sqrt{a^2 + b^2} = \sqrt{(400)^2 + (300)^2} = 500m$$

Solution

- Step 2:
- The V-C matrix Σ_L is:

$$\Sigma_{L(2,2)} = \begin{pmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ba} & \sigma_b^2 \end{pmatrix}$$

Solution

- **Step 2:**
- In general relative error for any quantity x

$$(re)_x = \frac{\sigma_x}{x}$$

$$\sigma_a = a \times (10 ppm) = (400m \times 1000mm) \times (10 \times 10^{-6}) = 4mm$$

$$\sigma_b = b \times (10 ppm) = (300m \times 1000mm) \times (10 \times 10^{-6}) = 3mm$$

Solution

- Step 2:

$$\text{covariance}(\sigma_{ab}) = \ell_{ab} \sigma_a \sigma_b = 0.5 \times 4 \times 3 = 6 \text{mm}^2$$

- By substituting in Σ_L we get:

$$\Sigma_{L(2,2)} = \begin{pmatrix} 16 & 6 \\ 6 & -9 \end{pmatrix} \text{mm}^2$$

Solution

- Step 3:

Here, the math. Model is non linear, thus we have to perform a partial derivatives to get J_L matrix

$$A = a \cdot b$$

$$c = \sqrt{a^2 + b^2}$$

- The Jacobian matrix J_L is:

$$J_{L(2,2)} = \begin{pmatrix} \frac{\partial A}{\partial a} & \frac{\partial A}{\partial b} \\ \frac{\partial c}{\partial a} & \frac{\partial c}{\partial b} \end{pmatrix}$$

Solution

- **Step 3:**

$$\frac{\partial A}{\partial a} = b = 300$$

m

$$\frac{\partial A}{\partial b} = a = 400$$

m

$$\frac{\partial c}{\partial a} = \frac{1}{c \cdot a} (2a) = \frac{a}{c} = \frac{400}{500} = 0.8$$

Unit less

$$\frac{\partial c}{\partial b} = \frac{1}{c \cdot b} (2b) = \frac{b}{c} = \frac{300}{500} = 0.6$$

Unit less

Solution

- Step 3:

$$J_{L(2,2)} = \begin{pmatrix} \frac{\partial A}{\partial a} & \frac{\partial A}{\partial b} \\ \frac{\partial c}{\partial a} & \frac{\partial c}{\partial b} \end{pmatrix} = \begin{pmatrix} 300m & 400m \\ 0.8 & 0.6 \end{pmatrix}$$

Before Scaling

Solution

- Step 4:
 - Scaling process.
 - As the unknown elements are different area in "m² and length in "m", each elements has to be treated individually.

Solution

- **Step 4:**

$$\sigma_A^2 = \left(\frac{\partial A}{\partial a}\right)^2 \sigma_a^2 + \left(\frac{\partial A}{\partial b}\right)^2 \sigma_b^2$$

$$m^4 = (m^2)(mm)^2 + (m^2)(mm)^2$$

• Each of the two terms on the RHS. needs to be divided by $(1000)^2$ to make all units m^4

• This is equivalent to dividing the first row by 1000

Solution

- **Step 4:**

$$\sigma_c^2 = \left(\frac{\partial c}{\partial a}\right)^2 \sigma_a^2 + \left(\frac{\partial c}{\partial b}\right)^2 \sigma_b^2$$

$$mm^2 = (unitless)^2 (mm)^2 + (unitless)^2 (mm)^2$$

- The second row doesn't require any scaling.

Solution

- Step 4:

$$J_{L_{(2,2)}} = \begin{pmatrix} \frac{300}{1000} & \frac{400}{1000} \\ 0.8 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.3 & 0.4 \\ 0.8 & 0.6 \end{pmatrix}$$

Solution

- Step 5:
Applying the covariance law to get Σ_x

$$X_{(2,1)} = \begin{pmatrix} A \\ c \end{pmatrix} \quad \Sigma_{X_{(2,2)}} = \begin{pmatrix} \sigma_A^2(m^4) & \sigma_{Ac}(m^2.mm) \\ \sigma_{cA}(mm.m^2) & \sigma_c^2(mm)^2 \end{pmatrix}$$

$$\Sigma_{X_{(2,2)}} = J_{L_{(2,2)}} \Sigma_{\bar{L}_{(2,2)}} J_{L_{(2,2)}}^T$$

Solution

- **Step 5:**

$$\sum X_{(2,2)} = \begin{pmatrix} 0.3 & 0.4 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix} \begin{pmatrix} 0.3 & 0.8 \\ 0.4 & 0.6 \end{pmatrix}$$

$$\sum X_{(2,2)} = \begin{pmatrix} 7.2 & 5.4 \\ 10.4 & 10.2 \end{pmatrix} \begin{pmatrix} 0.3 & 0.8 \\ 0.4 & 0.6 \end{pmatrix}$$

$$\sum X_{(2,2)} = \begin{pmatrix} 4.32m^4 & 9(mm.m^2) \\ 9(mm.m^2) & 19.24(mm)^2 \end{pmatrix}$$

Solution

- Step 5:

$$\sigma_A = \sqrt{4.32} = 2.08m^2$$

$$\sigma_c = \sqrt{19.24} = 4.39mm$$

$$\ell_{Ac} = \frac{\sigma_{Ac}}{\sigma_A \sigma_C} = \frac{9}{(2.08)(4.39)} = +0.987$$

Strong correlation “almost complete”

Solution

$$A = 120000m^2 \pm 2.08m^2$$

$$c = 500m \pm 4.39mm$$

Example 4

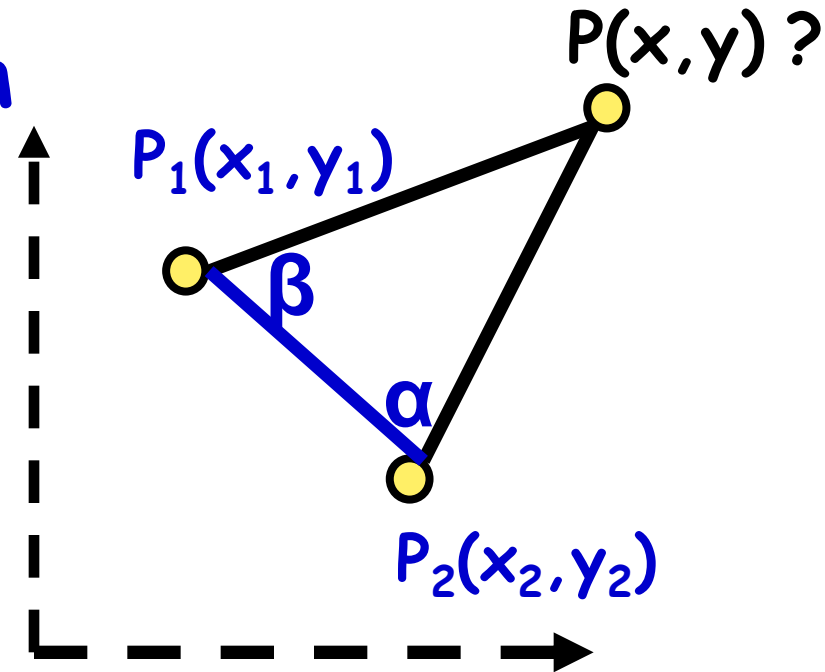
Given:

- $x_1 = 200 \text{ m}$, $y_1 = 500 \text{ m}$

$$\sum x_1 y_1 = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} cm^2$$

- $x_2 = 546.4 \text{ m}$,
 $y_2 = 300 \text{ m}$

$$\sum x_2 y_2 = \begin{pmatrix} 2 & -0.5 \\ -0.5 & 3 \end{pmatrix} cm^2$$



Example4

Given:

- $\alpha = 90^\circ$, $\sigma_\alpha = 3''$
- $\beta = 60^\circ$, $\sigma_\beta = 2''$

Considered uncorrelated

Required:

- x , y
- σ_x , σ_y
- Correlation between x , y

